

DONNA NOOK, United Kingdom

WMO: 033850

Lat: 53.475N

Lon: 0.153E

Elev: 26

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.6	(c) 30.7	(d) 23.2	(e) 17.5	(f) 32.6	(g) 25.3	(h) 19.4	(i) 33.7	(j) 39.9	(k) 41.6	(l) 35.8	(m) 42.0	(n) 10.1	(o) 290	(p) 0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.7	(c) 74.1	(d) 64.2	(e) 71.5	(f) 62.8	(g) 69.2	(h) 61.8	(i) 66.1	(j) 71.4	(k) 64.6	(l) 69.2	(m) 63.2	(n) 67.5	(o) 13.8	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.1	89.6	68.4	62.6	85.1	67.1	61.2	80.9	65.4	30.9	71.5	29.7	69.5	28.7	67.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 34.1	(b) 30.3	(c) 27.1	(e) 23.5	(f) 81.6	(g) 4.4	(h) 3.6	(i) 20.4	(j) 84.2	(k) 17.8	(l) 86.3	(m) 15.3	(n) 88.3	(o) 12.1	(p) 90.9		
(a) 34.1	(b) 30.3	(c) 27.1	(e) 23.2	(f) 70.4	(g) 3.8	(h) 3.4	(i) 20.4	(j) 72.8	(k) 18.2	(l) 74.8	(m) 16.0	(n) 76.7	(o) 13.2	(p) 79.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	40.4	41.2	43.5	47.0	51.7	56.9	60.9	61.2	57.5	52.3	45.5	41.2	
	DBStd	8.83	4.66	5.21	4.96	4.73	4.46	4.31	3.97	3.90	3.96	4.79	4.95	5.64	
	HDD50	1360	298	249	210	113	31	2	0	0	1	30	149	277	
	HDD65	5506	762	666	667	539	411	246	139	129	228	394	586	739	
	CDD50	1362	1	4	8	24	85	209	337	349	225	102	14	4	
	CDD65	29	0	0	0	0	0	3	11	12	2	1	0	0	
	CDH74	90	0	0	0	0	1	16	36	28	7	2	0	0	
	CDH80	7	0	0	0	0	0	2	2	3	0	0	0	0	
Wind	WSAvg	13.4	15.3	15.5	14.1	13.4	13.0	11.6	11.0	11.6	12.6	14.0	14.4	14.9	
Precipitation	PrecAvg	25.2	2.2	1.7	1.5	1.7	1.9	2.2	2.3	2.4	2.0	2.6	2.5	2.2	
	PrecMax	32.2	4.1	2.8	2.9	4.9	4.6	6.9	5.7	4.3	4.7	4.7	4.7	4.4	
	PrecMin	17.1	0.5	0.5	0.1	0.1	0.6	0.4	0.8	0.2	0.4	0.9	0.8	0.8	
	PrecStd	4.3	1.0	0.6	0.7	1.3	1.0	1.5	1.3	1.2	1.1	0.9	0.9	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	57.4	60.4	66.3	69.9	76.1	78.6	77.2	74.3	68.7	60.2	56.6	
		MCWB	51.9	52.9	53.5	57.0	60.5	64.9	66.0	65.9	64.2	61.8	56.8	53.3	
	2%	DB	52.3	54.1	56.6	61.2	65.7	71.0	73.9	73.4	70.1	64.0	57.1	54.4	
		MCWB	49.4	50.2	51.0	53.8	57.7	62.7	64.2	64.0	62.5	58.8	54.2	51.8	
	5%	DB	50.0	51.4	54.0	57.8	62.5	67.8	71.0	71.0	66.7	61.1	54.8	51.9	
		MCWB	47.4	48.1	49.3	51.8	55.6	60.7	62.9	62.4	60.4	57.4	52.2	49.6	
	10%	DB	47.8	49.2	51.6	55.0	59.7	64.9	68.5	68.7	64.2	59.1	52.7	49.4	
		MCWB	45.5	46.3	47.4	50.0	53.9	58.6	61.8	61.6	58.8	56.0	50.5	47.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	53.5	54.6	58.6	61.8	67.0	68.2	68.2	66.3	62.6	57.4	53.8
			MCDB	54.9	56.6	58.5	63.7	67.6	73.3	75.1	73.4	70.9	68.2	59.6	55.7
2%		WB	49.9	50.8	52.0	55.2	58.8	63.7	66.0	65.9	63.7	60.2	54.7	52.0	
		MCDB	52.0	53.5	55.6	59.6	64.0	69.4	71.5	70.7	67.8	62.9	56.6	54.1	
5%		WB	47.8	48.6	49.8	52.5	56.7	61.6	64.4	64.5	61.8	58.2	52.6	49.8	
		MCDB	49.7	51.1	53.3	56.6	61.1	66.4	68.8	68.8	65.1	60.2	54.4	51.7	
10%		WB	45.7	46.6	47.9	50.5	54.7	59.6	62.9	63.1	60.1	56.5	50.9	47.6	
		MCDB	47.5	48.9	51.0	54.1	58.6	63.8	67.2	66.9	63.0	58.6	52.4	49.3	
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.3	10.7	11.6	11.4	12.1	12.6	12.7	11.9	10.3	9.1	8.3	
		MCDBR	10.8	11.8	14.6	17.1	17.1	17.6	17.3	16.5	16.1	12.4	10.2	10.4	
		MCWBR	9.9	9.3	10.2	10.9	10.5	10.7	9.7	8.9	9.8	9.0	8.7	9.4	
	5% WB	MCDBR	10.6	11.8	14.0	15.8	15.3	16.3	15.1	13.7	13.9	11.7	9.8	10.4	
		MCWBR	10.0	9.7	10.5	11.1	10.3	10.9	9.5	8.8	9.7	9.3	8.8	9.6	
Clear-Sky Solar Irradiance	taub	0.328	0.344	0.375	0.402	0.400	0.407	0.409	0.398	0.377	0.357	0.329	0.316		
	taud	2.347	2.385	2.302	2.251	2.313	2.313	2.320	2.371	2.425	2.448	2.402	2.329		
	Ebn at Noon	201	239	254	262	269	268	265	261	252	231	200	181		
	Edh at Noon	19	24	32	38	38	38	38	34	29	23	18	16		
All-Sky Solar Radiation	RadAvg	205	424	802	1263	1591	1643	1593	1341	969	542	257	161		
	RadStd	20	58	77	132	116	125	140	89	82	40	25	16		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page